

Sec. 5 Experiments

Table 1: Headline cells, tail-500 over 5 seeds, decaying step-size. *full* sees every group every step. The fixed multiplier m/K diverges whenever observation rates are non-uniform.

Instance	GAVOT	unif. avg	m/K	full
Adult, prevalence ($\nu=.60$)	0.2269	0.2479	0.670	0.2073
Adult, aligned ($\nu=0$)	0.2075	0.2077	div.	0.2073
IMDb, skewed ($\nu=.31$)	86.06	91.63	8504	82.95
IMDb, aligned ($\nu=0$)	84.28	86.37	div.	82.95

1 EXPERIMENTS

Instances. *Adult* income classification [33]: logistic regression ($d=109$ with intercept, cross-entropy, $\eta=0.1$) on 100 race-homogeneous groups of 30 samples, the number of groups per race proportional to its prevalence (White/Black/Asian-Pac./Amer-Ind./Other = 85/10/3/1/1); the importance target is the fairness target itself, $1/5$ per race divided equally among its groups, so a group from the smallest race carries $p_i = 0.2$. *IMDb-Wiki* age regression [34]: linear regression of age on 128-d ResNet face embeddings ($\eta=10^{-2}$) over the first 100 identities with 30 images, with the cubic importance target $p_i \propto (101 - i)^3$. Observation rates are swept through a one-parameter family: on Adult $r \propto p^\beta$, $\beta \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, from uniform observation (prevalence regime, $\nu=.60$: the five groups of the three smallest races hold 60% of the target against $r_i \approx K/m$) to importance-aligned ($\nu=0$); on IMDb-Wiki $r_i \propto i^\beta$, $\beta \in \{0, \frac{1}{2}, 1, \frac{3}{2}, 2, 3\}$, from uniform observation ($\nu=.31$, nine groups) to the mirror image of p ($\nu=.88$, 41 groups), plus the aligned $r_i \propto 101 - i$ ($\nu=0$). Eleven instances, ν from 0 to .88.

Protocol. Each step observes $K=3$ groups drawn without replacement with probabilities $\propto r$, which induces q over the $\binom{100}{3}$ compositions; q is estimated from 10^6 draws of the sampler and the plan from (7) on the membership mask, once per instance. Each observed group takes $H=5$ local steps on its own data before the weighted average, the standard practical deviation from the one-step analysis, applied to every rule alike since all rules consume the same local updates from the same draw A^t . Rules: *GAVOT*; *group-blind averaging*, the uniform average over the observed groups (Proposition 2 with $\tilde{p}_i = r_i/K$); the *fixed multiplier* m/K , the rescaling $(m/K)p_i$ that presumes uniform observation; and *full*, every group every step weighted by p , the reference $\Phi(p)$. Stepsize $\eta_t = \eta/(1 + t/1000)$ unless stated, 4000 steps, 5 seeds; we report F_p averaged over the last 500 steps, mean $\pm$ std over seeds. On both instances $\arg \min F_w$ is available to machine precision (closed form for least squares, Newton for the logistic loss), so the floors $\Phi(p), \Phi(\tilde{p}), \Phi(\bar{p})$ of Theorem 8 are exact rather than estimated, and in every instance with $\nu = 0$ the iteration (7) converges to a row error below $3 \cdot 10^{-8}$, certifying the remaining inequalities of (5).

E1: the headline cells. Table 1 reports the four instances a practitioner would recognize. Transport wins

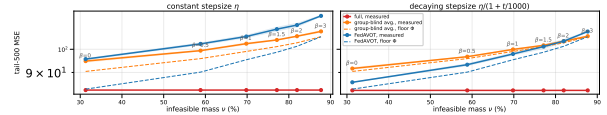


Figure 1: IMDb-Wiki severity sweep. Solid, measured tail-500 (± 1 std); dashed, the exact floor $\Phi(\cdot)$ each rule converges to. Left, constant stepsize: the optimization residual swamps the alignment gain and transport loses everywhere, even where the floors are seven units apart. Right, decaying stepsize: the residual collapses and the measured curves track their floors, so transport wins until the floors close at high ν . The sign is set by which of the two terms of (9) dominates, not by the regime label.

Table 2: IMDb-Wiki severity sweep, decaying stepsize. Δ_{floor} is the alignment gain of Theorem 8, *gain* the measured advantage of transport, t its Welch statistic (population std). The excess residual transport pays averages 1.9, and the sign flips exactly where Δ_{floor} crosses it. The five Adult skews all sit above their crossover and are omitted.

β	ν	Δ_{floor}	gain	t
0	.31	6.99	+5.57	20.7
0.5	.59	5.71	+3.46	11.3
1.0	.70	3.67	+2.05	5.4
1.5	.77	2.56	+0.58	1.1
2.0	.82	1.62	−0.03	−0.1
3.0	.88	0.02	−2.28	−7.9

on three: Adult under prevalence-proportional availability, 0.2269 against 0.2479 ($t=45$), and both IMDb cells, 86.06 against 91.63 ($t=21$) and 84.28 against 86.37 ($t=13$). On Adult with importance-aligned availability the two coincide at the oracle value, which is what Proposition 2 predicts when $\tilde{p} = p$: there is nothing to correct and transport does not manufacture a difference. The fixed multiplier diverges on both aligned instances and is far off on the other two, so naive rescaling is no alternative to solving (4).

E2: the crossover, and where the sign comes from. Table 2 and Fig. 1 are the paper’s main evidence. On Adult, whose five skews are omitted from the table, the measured gain tracks Δ_{floor} within 4% at three of five (.0134 against .0129, .0050 against .0049, .0002 against .0003) and exceeds it at the two most infeasible (.0210 against .0110, .0181 against .0144), group-blind averaging sitting further above its floor. On IMDb-Wiki the gain is uniformly below the floor by an excess optimization residual that is close to constant in β , mean 1.9 in MSE units, and the measured sign flips precisely where Δ_{floor} crosses that value, between $\beta=1.5$ (2.56, gain +0.58) and $\beta=2.0$ (1.62, gain −0.03). This is Theorem 8 read at finite T : the sign of the comparison is a quantity one computes in advance.

E3: the ℓ_1 bound is too loose to be the criterion.

The distances $\|p - \hat{p}\|_1$ and $\|p - \tilde{p}\|_1$ are also computed on every instance, and $\|p - \hat{p}\|_1 < \|p - \tilde{p}\|_1$ holds in all eleven, including the three where transport loses or ties. The ℓ_1 form of Theorem 8 therefore never predicts a loss and cannot serve as the criterion; the exact floors can.

E4: the residual is a stepsize artifact. The left panel of Fig. 1 runs the identical instances at constant stepsize. There transport loses on all six IMDB skews, by as much as 7.9, even at $\nu=.31$ where the floors are 6.99 apart. The transport weights $Y[\cdot, j]$ concentrate mass on rarely observed groups, so although Lemma 6 bounds the second moment identically for both rules, the variance about the mean is larger, and at constant stepsize the iterates sit on a noise floor proportional to it. Under $\eta_t = \eta/(1 + t/1000)$ that floor decays, the measured curves fall onto their dashed floors, and every sign reversal attributable to the stepsize disappears. The residual is a property of the schedule, not of the estimator.

E6: the decomposition in closed form. The mirrored IMDB-Wiki instance ($\nu=.88$) is where Theorem 8 can be read off exactly. The Sinkhorn limit truncates the target at the observation-rate ceiling, $\hat{p}_i \approx \min(p_i, r_i)$ up to renormalization (maximum deviation .023), with $\|p - \hat{p}\|_1 = 1.58$ of a possible 2. The frontier is least squares, so $\Phi(p) = 82.9$, matching the measured full plateau 82.95, and $\Phi(\hat{p}) = 105.9$: the alignment floor is 23.0 and the realized bias equals its exact expression $|(p - \hat{p})^\top L(\theta^\dagger)| = 30.7$, where L collects the per-group risks, while the ℓ_1 form $2M\|p - \hat{p}\|_1$ is conservative since $M \geq \max_i L_i(\theta^\dagger) \approx 454$. Group-blind averaging has the same floor to two decimals, $\Phi(\tilde{p}) = 105.9$: both surrogates starve the same groups, so Δ_{floor} vanishes and the entire measured gap between the two rules at this endpoint of Table 2 is residual, which is why it is the one instance where transport still loses by a significant margin under decay ($\beta=2$, where the floors are 1.6 apart, is a tie).

E5: risk aversion, and where it does not help. Sweeping (α, γ) on a 10×10 grid, the overall objective is minimized on the $\gamma = 1$ edge on every instance, which is $\kappa(1, \gamma) = 1$ made visible and confirms that risk aversion costs the average. The one place the CVaR layer earns its keep is the worst race on Adult under prevalence availability, where GAVOT+CVaR at $(0.2, 0.7)$ reaches 0.3624 against 0.3679 for GAVOT and 0.3994 for group-blind averaging. Per race, transport improves four of five (Black .179 vs .200, Asian-Pac .368 vs .399, Am-Ind .175 vs .189, Other .073 vs .119) and regresses only the majority group (.339 vs .332), which is what reweighting toward a category-uniform target is supposed to do.